

Therefore, the timing risk for a continuous strategy trading over a period of time t^* is:

$$\mathfrak{R} = \sqrt{X^2 \cdot P_0^2 \cdot t^* \sigma^2 \cdot \frac{1}{3}} \quad (7.45)$$

Substituting back for $t^* = \frac{1}{250} \cdot \frac{X}{ADV} \cdot \frac{1}{\alpha}$ we get:

$$\mathfrak{R}(\alpha) = \sqrt{X^2 \cdot P_0^2 \cdot \frac{1}{250} \cdot \frac{X}{ADV} \cdot \frac{1}{\alpha} \cdot \sigma^2 \cdot \frac{1}{3}} \quad (7.46)$$

Simplifying we have, timing risk for a single stock order:

$$\mathfrak{R}(\alpha) = \sigma \cdot X \cdot P_0 \sqrt{\frac{1}{250} \cdot \frac{1}{3} \cdot \frac{X}{ADV} \cdot \frac{1}{\alpha}} \quad (7.47)$$

QED

Timing Risk for a Basket of Stock

The timing risk for a basket of stock expressed in total dollars is:

$$TR_{\$}(x_k) = \sqrt{\sigma^2 \cdot \sum_{k=1}^n r'_k \tilde{C} r_k} \quad (7.48)$$

where,

r_k = column vector of unexecuted shares at the beginning of the period k

$$r_k = \begin{pmatrix} r_{1k} \\ r_{2k} \\ \vdots \\ r_{mk} \end{pmatrix}$$

r_{ik} = unexecuted shares of stock i at the beginning of period k

$\tilde{C}$ = covariance matrix expressed in terms of $(\$/\text{share})^2$ and scaled for a trading period.

To express the timing risk for the basket of stock in terms of basis points we simply divide the timing risk dollar amount by the initial value of the trade list $V_{\$} = \sum X \cdot P_0 \cdot 10^4$.

COMPARISON OF MARKET IMPACT ESTIMATES

Market impact parameters are computed for the different trading strategy representations of the model (Equation 7.13, Equation 7.14 and